

Alireza Salemi

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RESEARCH SUMMARY

PhD candidate and Google PhD Fellow working at the intersection of information retrieval and large language models, with research experience at Google DeepMind and Google Cloud AI.

Interests: RAG · LLM personalization · Post-training · Alignment · Agentic systems · Reasoning

EDUCATION

University of Massachusetts Amherst

Sep 2022 – Present

PhD, Computer Science (MSc 2024) · GPA 3.93/4.0 · advised by Hamed Zamani

- Awarded the **Google PhD Fellowship 2025** in Natural Language Processing.
- Two **SIGIR best-paper awards**: Best Short Paper (2024) and Best Student Paper (2023).
- **2nd place** (\$30K) at the 2026 Grounded Reasoning Cup, an AI-agent competition sponsored by Google, OpenAI, and Anthropic, ahead of teams from top universities.

University of Tehran

Sep 2017 – Feb 2022

BSc, Computer Engineering · GPA 19.60/20 (ranked 1st of 103) · advised by Azadeh Shakery

RESEARCH EXPERIENCE

Student Researcher — Google Cloud AI

Jun 2025 – Dec 2025

Hosts: Hamid Palangi and Mihir Parmar

- Designed blackboard-style multi-agent systems for autonomous information discovery in data-science workflows; 13–57% relative gains over the strongest baselines.

Student Researcher — Google DeepMind

Jun 2024 – May 2025

Hosts: Cheng Li, Mingyang Zhang, and Michael Bendersky

- Personalized and aligned LLMs with reinforcement learning, reinforced self-training, inference-compute scaling, reasoning, and pluralistic alignment (REST-PG, Pathways of Thoughts).

Data Science Intern — Lowe's

May 2023 – Aug 2023

Host: Surya Kallumadi

- Optimized dense retrievers that feed personal documents to LLMs via reinforcement learning and knowledge distillation from the model's own feedback, improving LLM personalization.

Research Assistant — Center for Intelligent Information Retrieval

Sep 2022 – Present

Advisor: Hamed Zamani

- Research on RAG, LLM personalization, and agentic systems—building benchmarks, reinforcement-learning and self-training methods, and evaluation frameworks grounded in retrieved evidence.

NLP Data Scientist — Sotoon

Nov 2021 – Aug 2022

Team lead: Hakimeh Fadaei

- Built an AutoML system that selects models and hyperparameters for text classification, powering sentiment analysis and automatic moderation of user-generated product reviews.

PUBLICATIONS

📄 1,693 citations | h-index 16 | i10-index 18 | *Google Scholar, July 2026*

SELECTED PUBLICATIONS

LaMP: When Large Language Models Meet Personalization

ACL 2024

Alireza Salemi, Sheshera Mysore, Michael Bendersky, Hamed Zamani

[Paper](#) · [Code](#)

First benchmark for LLM personalization (7 tasks); 577+ citations and a standard reference for the field.

Evaluating Retrieval Quality in Retrieval-Augmented Generation	SIGIR 2024
Alireza Salemi, Hamed Zamani 🏆 Best Short Paper	Paper · Code
<i>eRAG scores each retrieved document by its downstream contribution—Kendall’s τ +0.17–0.49 at up to 50× lower GPU cost.</i>	
A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering	SIGIR 2023
Alireza Salemi, Juan Altmayer Pizzorno, Hamed Zamani 🏆 Best Student Paper	Paper · Code
<i>DEDR encodes queries and documents in a shared space; +11.6–30.9% retrieval and state-of-the-art end-to-end VQA.</i>	
LLM-based Multi-Agent Blackboard System for Information Discovery in Data Science	Under review (NeurIPS 2026)
Alireza Salemi, Mihir Parmar, Palash Goyal, Yiwen Song, Jinsung Yoon, Hamed Zamani, Hamid Palangi, Tomas Pfister	Paper
<i>Specialized agents self-select tasks via a shared blackboard—no rigid hierarchy; 13–57% relative gains. (Google Cloud AI.)</i>	
Optimization Methods for Personalizing Large Language Models through Retrieval Augmentation	SIGIR 2024
Alireza Salemi, Surya Kallumadi, Hamed Zamani	Paper · Code
<i>First to optimize the retriever that feeds personal documents to an LLM via RL and knowledge distillation; significant gains on 6/7 LaMP tasks. (Lowe’s.)</i>	
GrepSeek: Training Search Agents for Direct Corpus Interaction	To be submitted (ICLR 2027)
Alireza Salemi, Chang Zeng, Atharva Nijasure, Jui-Hui Chung, Razieh Rahimi, Fernando Diaz, Hamed Zamani	Paper · Code
<i>Trains LLM agents to search corpora via shell commands using verified-trajectory generation + policy optimization (RL); strong gains across seven QA benchmarks.</i>	
OTHER PUBLICATIONS	
Critic-R: Improving Agentic Search using Instruction-tuned Retrievers with Natural Language Introspective Feedback	Under review (EMNLP 2026)
Md Zarif Ul Alam, Alireza Salemi, Hamed Zamani	Paper
Improving User Privacy in Personalized Generation: Client-Side Retrieval-Augmented Modification of Server-Side Generated Speculations	arXiv 2026
Alireza Salemi, Hamed Zamani	Paper · Code
Evaluation of Agents under Simulated AI Marketplace Dynamics	SIGIR 2026
To Eun Kim, Alireza Salemi, Hamed Zamani, Fernando Diaz	Paper
Learning to Reason for Multi-Step Retrieval of Personal Context in Personalized Question Answering	SIGIR 2026
Maryam Amirizani, Alireza Salemi, Hamed Zamani	Paper · Code
Learning from Natural Language Feedback for Personalized Question Answering	SIGIR 2026
Alireza Salemi, Hamed Zamani	Paper
Beyond a Million Tokens: Benchmarking and Enhancing Long-Term Memory in LLMs	ICLR 2026
Mohammad Tavakoli, Alireza Salemi, Carrie Ye, Mohamed Abdalla, Hamed Zamani, J. Ross Mitchell	Paper
Pathways of Thoughts: Multi-Directional Thinking for Long-Form Personalized Question Answering	WWW 2026
Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Zhuowan Li, Spurthi Amba Hombaiah, Weize Kong, Tao Chen, Hamed Zamani, Michael Bendersky	Paper
Plan-and-Refine: Diverse and Comprehensive Retrieval-Augmented Generation	ICTIR 2026
Alireza Salemi, Chris Samarinas, Hamed Zamani	Paper · Code
LaMP-QA: A Benchmark for Personalized Long-Form Question Answering	EMNLP 2025
Alireza Salemi, Hamed Zamani	Paper · Code

- Personalized Generation in the Large Model Era: A Survey** *ACL 2025*
 Yiyao Xu, Jinghao Zhang, Alireza Salemi, Xinting Hu, Wenjie Wang, Fuli Feng, Hamed Zamani, Xiangnan He, Tat-Seng Chua [Paper](#)
- ExPerT: Effective and Explainable Evaluation of Personalized Long-Form Text Generation** *Findings of ACL 2025*
 Alireza Salemi, Julian Killingback, Hamed Zamani [Paper](#) · [Code](#)
- Beyond Factual Accuracy: Evaluating Coverage of Diverse Factual Information in Long-Form Text Generation** *Findings of ACL 2025*
 Chris Samarinas, Alexander Krubner, Alireza Salemi, Youngwoo Kim, Hamed Zamani [Paper](#) · [Code](#)
- Accelerating Retrieval-Augmented Generation** *ASPLOS 2025*
 Derrick Quinn, Mohammad Nouri, Neel Patel, John Salihu, Alireza Salemi, Sukhan Lee, Hamed Zamani, Mohammad Alian [Paper](#)
- Reasoning-Enhanced Self-Training for Long-Form Personalized Text Generation** *arXiv 2025*
 Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Weize Kong, Tao Chen, Zhuowan Li, Michael Bendersky, Hamed Zamani [Paper](#)
- Learning to Rank for Multiple Retrieval-Augmented Models through Iterative Utility Maximization** *ICTIR 2025*
 Alireza Salemi, Hamed Zamani [Paper](#)
- Comparing Retrieval-Augmentation and Parameter-Efficient Fine-Tuning for Privacy-Preserving Personalization of Large Language Models** *ICTIR 2025*
 Alireza Salemi, Hamed Zamani [Paper](#) · [Code](#)
- CIIR@LiveRAG 2025: Optimizing Multi-Agent Retrieval-Augmented Generation through Self-Training** *LiveRAG @ SIGIR 2025*
 Alireza Salemi, Mukta Maddipatla, Hamed Zamani [Paper](#) · [Code](#)
- Towards a Search Engine for Machines: Unified Ranking for Multiple Retrieval-Augmented Large Language Models** *SIGIR 2024*
 Alireza Salemi, Hamed Zamani [Paper](#)
- LongLaMP: A Benchmark for Personalized Long-Form Text Generation** *Under review (TKDD)*
 Ishita Kumar, Snigdha Viswanathan, Sushrita Yerra, Alireza Salemi, Ryan Rossi, Franck Dernoncourt, Hanieh Deilamsalehy, Xiang Chen, Ruiyi Zhang, Shubham Agarwal, Nedim Lipka, Hamed Zamani [Paper](#)
- Retrieval-Enhanced Machine Learning: Synthesis and Opportunities** *Under review (IR Journal)*
 To Eun Kim, Alireza Salemi, Andrew Drozdov, Fernando Diaz, Hamed Zamani [Paper](#)
- Pre-Training Multi-Modal Dense Retrievers for Outside-Knowledge Visual Question Answering** *ICTIR 2023*
 Alireza Salemi, Mahta Rafiee, Hamed Zamani [Paper](#)
- PEACH: Pre-Training Sequence-to-Sequence Multilingual Models for Translation with Semi-Supervised Pseudo-Parallel Document Generation** *LoResMT @ EACL 2023*
 Alireza Salemi, Amirhossein Abaskohi, Sara Tavakoli, Yadollah Yaghoobzadeh, Azadeh Shakery [Paper](#) · [Code](#)
- ARMAN: Pre-training with Semantically Selecting and Reordering of Sentences for Persian Abstractive Summarization** *EMNLP 2021*
 Alireza Salemi, Emad Kebriaei, Ghazal Neisi Minaei, Azadeh Shakery [Paper](#) · [Code](#)
- A Comparative Analysis of Toxic Span Detection using Attention-based, Named Entity Recognition, and Ensemble Models** *SemEval @ ACL 2021*
 Alireza Salemi, Nazanin Sabri, Emad Kebriaei, Behnam Bahrak, Azadeh Shakery [Paper](#) · [Code](#)

HONORS AND AWARDS

Grounded Reasoning Cup — 2nd Place and \$30K prize: live AI-agent competition on retrieval and reasoning over enterprise documents (Data + AI Summit 2026); sponsored by Google, OpenAI, and Anthropic; finished ahead of teams from leading universities. 2026

Google PhD Fellowship in Natural Language Processing: awarded to exceptional PhD students with outstanding research potential; proposal on collaborative optimization of agents.	2025
PhD Candidacy with Distinction: for exceptional performance in the PhD comprehensive exam.	2025
SIGIR 2024 Best Short Paper Award: Evaluating Retrieval Quality in Retrieval-Augmented Generation.	2024
SIGIR 2023 Best Student Paper Award: A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering.	2023
W. Bruce Croft Graduate Scholarship: for research in information retrieval and NLP.	2023, 2025
UMass Amherst CICS Scholarship: merit-based award for selected entering graduate students.	2022
Best Thesis Award, Computer Engineering: University of Tehran — BSc thesis selected as the best in the Computer Engineering graduating class for its research contribution.	2021
University of Tehran top-student awards: UT Sponsors Foundation Top Student, UT Scholarship, and Faculty of Engineering Award (top of 103 students).	2017–2020

INVITED TALKS

Personalization in the LLMs Era — Bocconi University	May 2026
Personalization in the LLMs Era — Carnegie Mellon University	Oct 2025
Personalization in the LLMs Era — Google DeepMind	May 2025
Personalization in the LLMs Era — UMass Amherst	Apr 2025
Knowledge-Intensive Visual Question Answering — QIAU	Dec 2023
When Large Language Models Meet Personalization — UMass ML Club	Oct 2023
When Large Language Models Meet Personalization — Google Research	May 2023

PROFESSIONAL SERVICE

Organizing

- **Workshop organizer**, Second Workshop on Customizable NLP (CustomNLP4U) — ACL 2026.
- **Tutorial organizer**, Retrieval-Enhanced Machine Learning — SIGIR 2025 and SIGIR-AP 2024.

Program Committee and Reviewing

- **Program committee and reviewer** (80+ papers): NeurIPS, ACL, EMNLP, SIGIR, WWW, CIKM, ICTIR, SIGIR-AP, LLM-IGS; journals ACM TOIS and IEEE TKDE.

Mentoring

- Independent study — mentored 15+ students across RAG, personalization, and agents.
- Industry Mentorship Program — MS students on LLM personalization, with Adobe mentors.
- Early Research Scholar Program — undergraduates on QA and LLM personalization.